



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
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Seat No.						
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT,
PUNE
Programme: (020343) MASTER OF BUSINESS ADMINISTRATION (BUSINESS ANALYTICS)

Batch: 2024-26
Semester: I
Course: Quantitative Methods
Course Code: 0203430102

Date: 13/03/2025
Day: Thursday

Maximum Marks: 60
Time: 09:30 - 12:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions :

- 1) All questions are compulsory
- 2) Use of calculators is allowed.

Q.1 A well known hospital in Hinjawadi area of Pune is organizing a blood donation and health check up camp on Independence Day, for corporate employees in this region. The hospital wishes to employ volunteers to staff the reception desk between 8:00 A.M. and 10:00 P.M. Each volunteer works three consecutive hours except for those starting at 8:00 P.M. who work for two hours only. The minimum need for volunteers is approximated by a step function over 2-hour intervals starting at 8:00 A.M. as :

Time slot	8 to 10 AM	10 AM to 12 PM	12 to 2 PM	2 to 4 PM	4 to 6 PM	6 to 8 PM	8 to 10 PM
Minimum volunteers needed	4	6	8	6	4	6	8

Because most volunteers are retired individuals, they are willing to offer their services at any hour of the day (8:00 A.M. to 10:00 P.M.). However, because of the large number of charities competing for their service, the number needed must be kept as low as possible. Determine an optimal schedule for the start time of the volunteers.

Formulate the Linear Programming model.

(CO1)(10 Marks)

Q.2 MotoBiz car wash facility operates with only one bay. Cars arrive according to a Poisson distribution with a mean of one car per 15 min, and may wait in the facility's parking lot if the bay is busy. The time for washing and cleaning a car is exponential, with a mean of 10 minutes. Cars that cannot park in the lot can wait in the street bordering the wash facility. This means that, for all practical purposes, there is no limit on the size of the system. The manager of the facility wants to determine: (CO1)

- Probability that the system is idle (2 marks)
- Average expected time for the car in the total washing process (2 marks)
- Expected number of cars waiting for wash (2 marks)
- The owner has a space for three waiting cars as allocated parking lot. Additional cars can be parked by the side of the road. What is the probability that the owner need not expand the current parking facility? (4 marks)

Q.3 Five workers are available to work on six machines. The respective costs associated with each worker-machine assignment are given below. Due to limitations of the skills, some worker-machine combinations are restricted. Determine the optimal assignment and total minimum cost of assignment. (CO1) (8 Marks)

	M1	M2	M3	M4	M5	M6
W1	12	3	6	-	5	9
W2	4	11	-	5	-	8
W3	8	2	10	9	7	5
W4	-	7	8	6	12	10
W5	5	8	9	4	6	1

Q.4 One of the main products of the P & T COMPANY is canned peas. The peas are prepared at three canneries (near Bellingham, Washington; Eugene, Oregon; and Albert Lea, Minnesota) and then shipped by truck to five distributing warehouses in the western United States (Sacramento, California; Salt Lake City, Utah; Rapid City, South Dakota; and two at Albuquerque, New Mexico). Because the shipping costs are a major expense, management is initiating a study to reduce them as much as possible. For the upcoming season, an estimate has been made of the output from each cannery, and each warehouse has been allocated a certain amount from the total supply of peas. This information (in units of truckloads), along with the shipping cost per truckload for each cannery-warehouse combination is given below. Thus, there are a total of 300 truckloads to be shipped. The problem now is to determine which plan for assigning these shipments to the various cannery-warehouse combinations would minimize the total shipping cost.

(CO1) (12 Marks)

	WH1	WH2	WH3	WH4	WH5	Supply
Cannery 1	1	9	13	36	51	50
Cannery 2	24	12	16	20	1	100
Cannery 3	14	33	1	23	26	150
Demand	100	70	50	40	40	

- Q.5 A) A businessman has four alternative actions (A, B, C, D) and each of these follow three possible events. The conditional payoffs for each action event combination are as under.

Event s	Actions			
	A	B	C	D
I	40	0	-50	30
II	-20	60	90	10
III	70	30	20	40

If the probabilities of occurrence of three events (I, II, III) are 0.3, 0.5 and 0.2 respectively, find the optimal decision under risk and value of perfect information.

(CO1)(5 Marks)

- B) A firm needs 1,000 electric drills per year. The ordering cost for these is \$100 per order and the carrying cost is assumed to be 40% of the unit cost. In orders of less than 120, drills cost \$78; for orders of 120 or more, the cost drops to \$50 per unit. Should we take advantage of the quantity discount?

(CO1) (5 Marks)

Q.6 The following tables provide sensitivity analysis reports for an LP problem. Analyse the report and answer the questions below.

(CO1)(10 marks)

Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
Model 1	0	-59.57142857	69	59.57142857	1E+30
Model 2	10	0	59	26.71428571	1E+30
Model 3	94.85714286	0	100	1E+30	13.37500001
Model 4	0	-15.2857143	98.99999999	15.2857143	1E+30

Constraints

Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
MS	219.7142857	0	900	1E+30	680.2857143
Assembly	724	14.28571429	724	1	664
Paint	848.8571429	0	850	1E+30	1.142857143
MX	10	-26.71428571	10	0.533333333	10

- What is the value of the objective function in the optimal solution? What is the nature of objective – minimising or maximising? Why?
- The Assembly constraint is reduced to 704, what will be the resultant change in objective function value?
- What kind (less than or greater than) of constraint is MX? If MX constraint RHS is changed to 8, what will be the resultant change in the objective function value?
- The optimal solution suggests that no units of Model 4 are to be produced. What should be the objective coefficient of Model 4 to make it profitable?
- If the Paint constraint is decreased to 848 units, what will be the resultant change in objective function value?